

MLFF MLCV-AGG1 conventional cross-validation specification

Status: implemented in mdstats 0.20.136a0 (MLCV-AGG1)

Purpose

MLCV-AGG1 converts frozen per-run MLCV-SELECT1 fold representatives into conventional cross-validation evidence. The outer CV fold is evaluation-only evidence. It cannot stop training, rank epochs, choose among the top-K shortlist, or replace a representative after seeing its result.

Fold evaluation contract

For fold i , SELECT1 first freezes one representative using only the nested checkpoint-selection domains $V_{i_light} \rightarrow V_{i_full}$ plus independent TRUE_DFT replay validation. Only then does AGG1 expose the complete outer target fold C_i .

AGG1 evaluates exactly the frozen representative checkpoint on C_i . If SELECT1 produced no representative, the fold records an explicit **no_representative** failure and no outer-fold model inference is purchased. If the representative exists, an outer target force-component RMSE is computed and compared with the configured authoritative target ceiling.

An outer result may therefore produce only:

- **passed**;
- **no_representative**; or
- **outer_target_threshold_exceeded**.

It cannot trigger fallback to another epoch or checkpoint from the same fold run.

Replay semantics

Replay is not a rotating CV domain. AGG1 does not re-infer replay and does not create fold-specific replay partitions. It reuses the frozen representative's authenticated complete TRUE_DFT R_{full} absolute replay RMSE, matched foundation $R0_{full}$, and signed replay degradation $\Delta R_{full} = R_{full} - R0_{full}$ from MLCV-SELECT1.

The deployment-oriented combined fold score is

$$S_i = (w_T T_{outer,i} + w_R \Delta R_{full,i}) / (w_T + w_R).$$

This combined score is useful operational evidence but does not change the interpretation of $T_{outer,i}$ as the statistically clean rotating target CV quantity.

Per-seed aggregation

For every configured seed, all configured folds must be represented exactly once. All-fold survival is a hard robustness requirement: every fold must have a SELECT1 representative and every outer target RMSE must satisfy the configured target ceiling.

For target, replay degradation, absolute replay RMSE, and combined score separately, AGG1 reports:

- mean;
- sample standard deviation;
- minimum;
- maximum;
- range; and
- worst fold.

The sample standard deviation uses the conventional $N-1$ denominator when at least two folds are available. Cross-fold dispersion is diagnostic-only in AGG1 v1. No hard standard-deviation or range cutoff is introduced without empirical calibration and a future identity-bearing policy revision.

Production authority

Fold representatives are permanently production-ineligible. AGG1 creates no production model selection and no committee. It publishes CV robustness evidence only. Production comparison across optimizer seeds belongs to MLCV-FINAL1 and may consume only final-development representatives.

Zero-fold campaigns

A user may intentionally configure zero CV folds. Such a variant is recorded as `cv_not_performed`, not as conventional robustness evidence. It does not manufacture fold statistics.

Restart and immutability

Outer-fold evaluation records are bound to the campaign run identity, SELECT1 record digest, representative checkpoint SHA/epoch, outer artifact digest/SHA, and AGG1 policy digest. Re-running evaluation may reuse authenticated prediction/evaluation records, but any attempt to bind different outer evidence or a different checkpoint to the same immutable fold record fails closed.

0.20.140a0 replay-degradation correction

AGG1 combined scores use the representative's authenticated signed `DeltaR_full`, never raw absolute `R_full`. Absolute replay RMSE remains

a separately reported physical diagnostic. Historical aggregate schemas keep their original raw-replay meaning and are stale for current FINAL1 authority.